

MODULE 07 · DOMAIN 3 - DESIGN HIGH-PERFORMING ARCHITECTURES

Compute, Containers and Serverless

Choosing where the code runs.

DOMAIN 3 of 4 EXAM WEIGHT 24% MODULE 7 of 22

The exam has a preference order:
serverless, then containers, then instances.
Deviate only when the question gives you a reason.

The compute options, and their exam signals

OPTION	SIGNAL IN THE QUESTION
AWS Lambda	Event-driven, short tasks, unpredictable or spiky traffic, no servers to manage
AWS Fargate	Containers, but no cluster capacity to manage
Amazon ECS on EC2	Containers where you need control of the hosts, or want Spot or Reserved pricing
Amazon EKS	Kubernetes specifically, or existing Kubernetes skills and tooling
Amazon EC2	Specific OS, licensing, long-running processes, or full control
AWS Batch	Large numbers of batch jobs with queuing and dependencies
AWS Elastic Beanstalk	Deploy a standard web application without designing the infrastructure

ON THE EXAM

The word Kubernetes means EKS. It is the one service the exam identifies by name rather than by need.

Lambda limits that decide questions

- ◆ **Fifteen minute maximum execution time.** A hard limit. No support request changes it.
- ◆ **Memory from 128 MB to 10 GB,** and CPU scales with memory. More memory can mean lower total cost.
- ◆ **Ephemeral /tmp storage** from 512 MB up to 10 GB.
- ◆ **Cold starts** matter for latency-sensitive work. Provisioned concurrency keeps environments warm.
- ◆ **Reserved concurrency** caps a function, protecting downstream systems from being overwhelmed.

ON THE EXAM

'The job takes twenty minutes' rules out Lambda immediately. That single number appears in question after question.

Fargate or EC2 launch type?

FARGATE

No hosts to manage

- AWS runs the underlying capacity
- Pay per task vCPU and memory
- Scales without cluster capacity planning
- Usually correct when overhead is the qualifier

ECS ON EC2

You manage the hosts

- You patch, scale and secure the instances
- Spot and Reserved pricing available
- Needed for GPUs or special instance types
- Cheaper at steady high utilisation

THE TIEBREAKER

LEAST operational overhead means Fargate. MOST cost-effective at steady heavy load can mean EC2 with Spot or Reserved.

Placement groups and specialised compute

- ◆ **Cluster placement group.** Instances packed close together for lowest latency and highest throughput. HPC workloads.
- ◆ **Spread placement group.** Instances on distinct hardware. For a small number of critical instances.
- ◆ **Partition placement group.** Groups on separate racks. For large distributed systems such as HDFS or Cassandra.
- ◆ **Elastic Fabric Adapter.** Network interface for tightly coupled HPC and machine learning.
- ◆ **Graviton.** AWS processors offering better price performance for compatible workloads.

ON THE EXAM

'Tightly coupled, node-to-node communication, lowest latency' means a cluster placement group, often with EFA.

When an image is uploaded to S3, a function must create three thumbnail sizes. Uploads are unpredictable: some hours see none, and occasionally several thousand arrive in a minute. Each conversion takes about eight seconds.

The current design runs two EC2 instances continuously, polling the bucket. They are idle most of the time and overwhelmed during bursts.

What is the MOST cost-effective solution?

- ☐ A An Auto Scaling group scaling on the number of objects in the bucket
- ☐ B Lambda triggered by S3 event notifications
- ☐ C An ECS service on EC2 with scheduled scaling
- ☐ D Larger EC2 instances to absorb the bursts

◆ PAUSE AND DECIDE

What is the MOST cost-effective solution?

- A

 Auto Scaling on object count
Still pays for running instances during idle hours, and scaling takes minutes while the burst lasts one.
- B

 Lambda on S3 events
Nothing runs when nothing is uploaded, thousands of invocations run in parallel during a burst, and eight seconds is far inside the limit.
- C

 ECS with scheduled scaling
Scheduled scaling needs a predictable pattern, and the question states the load is unpredictable.
- D

 Larger instances
Pays more per hour to sit idle most of the time, and a fixed size still has a burst ceiling.

THE SIGNAL

Event-driven, spiky, short duration, and idle much of the time is the definitive Lambda signature.

A nightly financial reconciliation process reads twelve million records, performs calculations, and writes results. It currently runs on an EC2 instance and takes about ninety minutes.

The team wants to move it to a serverless architecture to avoid running the instance all day, and proposes rewriting it as a Lambda function.

How should the solutions architect respond?

- ☐ A Proceed with Lambda, increasing memory to speed up execution
- ☐ B Use AWS Batch or Fargate on a schedule, because the runtime exceeds Lambda's limit
- ☐ C Split the work across Lambda functions invoked in a loop until complete
- ☐ D Keep the EC2 instance but stop it outside the processing window

◆ PAUSE AND DECIDE

How should the solutions architect respond?

- A

Lambda with more memory
More memory speeds execution but cannot cross the fifteen-minute wall. Ninety minutes will not compress six-fold reliably.
- B

Batch or scheduled Fargate task
Serverless in the sense that matters, with no runtime ceiling and nothing running between executions.
- C

Chained Lambda functions
Technically possible but adds checkpointing and orchestration complexity to work around a limit that another service does not have.
- D

Stop the instance when idle
A reasonable cost saving, but it keeps the instance to patch and manage and ignores the stated goal.

THE SIGNAL

Serverless does not mean Lambda. Fargate and Batch are serverless too, and neither has a fifteen-minute ceiling.

A company runs twenty microservices in Docker containers on a self-managed cluster of EC2 instances. The platform team spends most of its time patching hosts, managing cluster capacity and handling scaling.

The services have steady but modest traffic. The company wants to reduce the platform team's workload and does not use Kubernetes.

Which solution meets this with the LEAST operational overhead?

- ☐ A Amazon EKS with a managed node group
- ☐ B Amazon ECS with the Fargate launch type
- ☐ C Amazon ECS on EC2 with Auto Scaling
- ☐ D Migrate the containers to Lambda container images

◆ PAUSE AND DECIDE

Which solution meets this with the LEAST operational overhead?

- A

EKS with managed node groups
Introduces Kubernetes to a team that does not use it, adding a control plane charge and a large body of new complexity.
- B

ECS with Fargate
No hosts to patch, no cluster capacity to plan, and containers run unchanged. Directly removes the team's stated workload.
- C

ECS on EC2 with Auto Scaling
Automates scaling but leaves patching and capacity management, which is most of what the team currently does.
- D

Lambda container images
Lambda can run container images, but converting twenty long-running services to an event-driven model is a rewrite, not a migration.

THE SIGNAL

Containers plus LEAST operational overhead plus no mention of Kubernetes equals ECS on Fargate.

A research team runs molecular simulations across sixty-four EC2 instances that exchange data with each other continuously throughout each run. Runs take four to six hours.

The team reports that performance is far below expectations and that the bottleneck appears to be network latency between the nodes rather than processor speed.

Which change would MOST improve performance?

- ☐ A Launch the instances in a cluster placement group with Elastic Fabric Adapter
- ☐ B Spread the instances across three Availability Zones for resilience
- ☐ C Move the instances into a spread placement group
- ☐ D Attach Provisioned IOPS EBS volumes to each instance

◆ PAUSE AND DECIDE

Which change would MOST improve performance?

- A

 Cluster placement group with EFA
Packs nodes onto the same low-latency network segment, and EFA provides the high-throughput interconnect these workloads need.
- B

 Spread across three AZs
Improves resilience and makes latency worse, since cross-zone traffic travels further. It optimises the wrong property.
- C

 Spread placement group
Deliberately separates instances onto distinct hardware, which is the opposite of what tightly coupled nodes need.
- D

 Provisioned IOPS volumes
Addresses disk throughput. The question states the bottleneck is network latency between nodes.

THE SIGNAL

Nodes talking to each other constantly means cluster placement group. Resilience answers make latency worse here.

A Lambda function reads from an SQS queue and writes to a small Amazon RDS instance. During busy periods Lambda scales to over a thousand concurrent executions, exhausting the database connection pool and causing widespread failures. The database cannot be resized in the near term.

Which change addresses this MOST directly?

- ☐ A Set reserved concurrency on the Lambda function to a level the database can sustain
- ☐ B Set provisioned concurrency on the Lambda function
- ☐ C Increase the Lambda function's memory allocation
- ☐ D Increase the SQS visibility timeout

◆ PAUSE AND DECIDE

Which change addresses this MOST directly?

- A

 Reserved concurrency
Caps how many copies run at once, so the database never sees more connections than it can handle. Messages simply wait in the queue.
- B

 Provisioned concurrency
Keeps environments warm to remove cold starts. It addresses latency, not the number of concurrent connections.
- C

 More memory
Makes each invocation faster. Faster invocations open connections sooner, which can make the problem worse.
- D

 Longer visibility timeout
Changes how long a message stays hidden. It does not limit how many functions run at once.

THE SIGNAL

Reserved concurrency limits. Provisioned concurrency warms. The names are similar and the exam exploits that.

A company must run a legacy application that requires a specific version of Windows Server and is licensed per physical socket, with the licence terms requiring dedicated physical hardware.

The team suggests containerising it on Fargate to reduce management overhead.

What should the solutions architect recommend?

- ☐ A Fargate, since it removes host management
- ☐ B EC2 Dedicated Hosts, which provide visibility of physical sockets for licensing
- ☐ C EC2 Dedicated Instances on shared hardware
- ☐ D Lambda with a custom Windows runtime

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Fargate

Removes overhead but gives no visibility or control of physical hardware, so socket-based licensing cannot be satisfied.

C Dedicated Instances

Isolates your instances from other customers but does not expose the physical socket and core detail licensing needs.

B Dedicated Hosts

Gives you a physical server with visible sockets and cores, which is what socket-based and bring-your-own licensing requires.

D Lambda with Windows runtime

Lambda does not offer Windows runtimes, and a long-running licensed server application is not an event-driven function.

THE SIGNAL

Per-socket or per-core licensing that requires dedicated hardware means Dedicated Hosts. This is the one licensing signal worth memorising.

A genomics company submits around forty thousand independent analysis jobs each week. Each takes between ten minutes and six hours, some require GPU instances, and some must not start until an earlier job has finished.

The team currently manages queuing and instance provisioning with custom scripts, which frequently leave instances running after jobs complete.

Which service is designed for this workload?

- ☐ A AWS Batch
- ☐ B AWS Lambda with SQS
- ☐ C Amazon EMR
- ☐ D AWS Step Functions with EC2 instances

◆ PAUSE AND DECIDE

Which service is designed for this workload?

- A** AWS Batch
Purpose-built for large volumes of batch jobs: job queues, dependencies between jobs, GPU support, and automatic provisioning and termination of compute.
- B** Lambda with SQS
Jobs run up to six hours, far past the fifteen-minute ceiling, and Lambda offers no GPU.
- C** Amazon EMR
A managed big data framework for Spark and Hadoop. These are independent jobs, not a distributed data-processing cluster.
- D** Step Functions with EC2
Orchestrates workflows but does not manage a job queue or provision and terminate batch compute. You would rebuild much of Batch.

THE SIGNAL

Job queues, job dependencies and automatic compute provisioning together mean AWS Batch. Batch and EMR are not interchangeable.

A small development team has built a standard three-tier Python web application. They need it running on AWS with load balancing, Auto Scaling, health monitoring and a straightforward deployment process.

Nobody on the team has infrastructure experience, and there is no dedicated operations function. They want to spend their time on the application.

Which approach is MOST appropriate?

- ☐ A Write CloudFormation templates defining the VPC, ALB, Auto Scaling group and instances
- ☐ B Deploy the application with AWS Elastic Beanstalk
- ☐ C Build a Kubernetes cluster on EKS with managed node groups
- ☐ D Launch EC2 instances manually and configure the tiers by hand

◆ PAUSE AND DECIDE

Which approach is MOST appropriate?

A CloudFormation templates

Gives full control and repeatability, but requires exactly the infrastructure expertise the team does not have.

C EKS cluster

The highest-complexity option on the list, for a team with no infrastructure experience and a standard web application.

B Elastic Beanstalk

You upload the application and it provisions the load balancer, Auto Scaling, monitoring and deployment pipeline for you, using standard AWS resources you can still access.

D Manual EC2 configuration

No Auto Scaling, no repeatability, and the most ongoing work of any option.

THE SIGNAL

Elastic Beanstalk has one signal: a standard application and a team that does not want to design infrastructure.

Module 7 takeaways

- ◆ **Prefer serverless, then containers, then instances**, unless the question gives a reason not to.
- ◆ **Fifteen minutes** is the number that eliminates Lambda. Serverless still includes Fargate and Batch.
- ◆ **Reserved concurrency limits. Provisioned concurrency warms.**
- ◆ **Nodes talking to each other** means cluster placement group, often with EFA.
- ◆ **Per-socket licensing** means Dedicated Hosts, not Dedicated Instances.

ON THE EXAM

Next: storage and database performance, where the tiebreakers are about access patterns rather than services.